

5.5 Properties of the estimator under second-moment assumptions

Least squares was defined for a fixed data vector. Now let the data be random, generated by the linear model of Definition 5.3.1:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

with $\mathbf{X}$ of full rank p . The estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is then a random vector, and so are $\hat{\mathbf{Y}}$ and $\hat{\boldsymbol{\varepsilon}}$. This section computes their means and covariances. Only the second-moment assumptions are used, so the results hold for skewed, heavy-tailed or discrete errors alike, as long as they have finite variance.

5.5.1 Mean and covariance of the estimator

The key observation is that $\hat{\boldsymbol{\beta}}$ is a *linear* function of $\mathbf{Y}$: $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$ with the constant matrix $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. The moments of a linear transformation were worked out in [Chapter 2](#).

Theorem 5.5.1. Moments of the least squares estimator

Let $\mathbf{X}$ have full rank p and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$.

- (a) If $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, then $\mathbf{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$: the least squares estimator is unbiased.
- (b) If moreover $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$, then

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (5.5.1)$$

Proof. Let $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$. By Theorem 2.2.1(a), $\mathbf{E}(\hat{\boldsymbol{\beta}}) = \mathbf{A} \mathbf{E}(\mathbf{Y}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}$. By Theorem 2.2.1(b),

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \mathbf{A}(\sigma^2 \mathbf{I}_n) \mathbf{A}^\top = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1},$$

using the symmetry of $(\mathbf{X}^\top \mathbf{X})^{-1}$. ■

The proof is two lines, but the division of labour between the assumptions is worth stating. Unbiasedness needs only (L1): the errors may have unequal variances or be correlated, and $\hat{\boldsymbol{\beta}}$ still centres on $\boldsymbol{\beta}$. The covariance formula needs (L2) and (L3) as well. Neither needs normality. Under normality $\hat{\boldsymbol{\beta}}$ is itself normal, since it is a linear function of a normal vector (Theorem 3.3.1), so $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$. [Chapter 7](#) builds exact inference on this (Theorem 7.5.1).

The matrix $(\mathbf{X}^\top \mathbf{X})^{-1}$ carries all the information about how the design affects precision. Its diagonal entries give the variances of the individual coefficients, and its off-diagonal entries their covariances. When two columns are strongly positively correlated, their coefficients are typically estimated with a strong *negative* correlation: the data can tell how large the two effects are together, but not how to split the total between them.

Corollary 5.5.2. Linear functions of the coefficients

Under the assumptions of [Theorem 5.5.1\(b\)](#), let $\mathbf{a} \in \mathbb{R}^p$ and let $\mathbf{L}$ be a constant $q \times p$ matrix. Then

$$E(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \mathbf{a}^\top \boldsymbol{\beta}, \quad \text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a},$$

and $\text{Cov}(\mathbf{L} \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{L} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{L}^\top$. In particular, the estimated mean response at a design point $\mathbf{x}_0$, $\hat{\mu}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$, is unbiased for $\mu_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$ with $\text{Var}(\hat{\mu}_0) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$, and $\text{Var}(\hat{\beta}_j) = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$.

Proof. Apply Theorem 2.2.1 to $\mathbf{L} \hat{\boldsymbol{\beta}}$, with covariance $\mathbf{L} \text{Cov}(\hat{\boldsymbol{\beta}}) \mathbf{L}^\top$, and take $\mathbf{L} = \mathbf{a}^\top$, $\mathbf{L} = \mathbf{x}_0^\top$ and $\mathbf{L} = \mathbf{e}_j^\top$. ■

Linear functions of $\boldsymbol{\beta}$ are what applications usually need: a single coefficient, the difference between two coefficients, the mean response at a new design point, or the change in mean response between two profiles. [Chapter 7](#) shows that for every such function, $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$ has the smallest variance among all linear unbiased estimators (the Gauss–Markov theorem, Theorem 7.2.1).

5.5.2 Fitted values and residuals

Proposition 5.5.3. Moments of fitted values and residuals

Under the linear model with $\mathbf{X}$ of full rank and $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$:

- (a) $E(\hat{\mathbf{Y}}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}$;
- (b) $E(\hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2 (\mathbf{I} - \mathbf{H})$; in particular $\text{Var}(\hat{\varepsilon}_i) = \sigma^2 (1 - h_{ii})$, where h_{ii} is the i th diagonal entry of $\mathbf{H}$;
- (c) $\text{Cov}(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$;
- (d) $0 \leq h_{ii} \leq 1$ for every i , and $\sum_i h_{ii} = p$.

Proof. $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ and $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$, with $\mathbf{H}$ symmetric and idempotent and $\mathbf{H}\mathbf{X} = \mathbf{X}$ (Proposition 5.4.2(a)). By Theorem 2.2.1, $E(\hat{\mathbf{Y}}) = \mathbf{H}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}\mathbf{H}^\top = \sigma^2 \mathbf{H}$; similarly $E(\hat{\boldsymbol{\varepsilon}}) = (\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2 (\mathbf{I} - \mathbf{H})^2 = \sigma^2 (\mathbf{I} - \mathbf{H})$. The cross-covariances are $\text{Cov}(\mathbf{H}\mathbf{Y}, (\mathbf{I} - \mathbf{H})\mathbf{Y}) = \sigma^2 \mathbf{H}(\mathbf{I} - \mathbf{H}) = \mathbf{0}$ and $\text{Cov}(\mathbf{A}\mathbf{Y}, (\mathbf{I} - \mathbf{H})\mathbf{Y}) = \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{H}) = \sigma^2 (\mathbf{A} - \mathbf{A}\mathbf{H}) = \mathbf{0}$, since $\mathbf{A}\mathbf{H} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{A}$. For (d), $\sigma^2 h_{ii}$ and $\sigma^2 (1 - h_{ii})$ are variances, so both are nonnegative, and $\sum_i h_{ii} = \text{tr } \mathbf{H} = p$. ■

Two consequences deserve emphasis. First, residuals are *not* uncorrelated and *not* of equal variance, even when the errors are both. The residual of a case with h_{ii} near 1 has variance near zero: the fit is pulled so close to such a case that its residual says little about its error. The number h_{ii} is the **leverage** of case i ([Section 6.8](#), [Chapter 20](#)). Second, the fitted values and the residuals are uncorrelated. Under normality they are then independent (Theorem 3.4.1), which is the key to the exact tests of [Chapter 11](#).

5.5.3 The straight line

Specializing to $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$, the inverse Eq. 5.4.4 gives the classical formulas at once.

Corollary 5.5.4. Moments for a straight line

In the model Eq. 5.2.1 with $S_{xx} > 0$,

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \quad \text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}},$$

and the estimated mean response at x_0 has

$$\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right). \quad (5.5.2)$$

Both estimates are unbiased, and $\bar{Y}$ is uncorrelated with $\hat{\beta}_1$.

Proof. The first three are the entries of $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ in Eq. 5.4.4. For the mean response, Corollary 5.5.2 with $\mathbf{x}_0 = (1, x_0)^\top$ gives $\sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} - \frac{2x_0\bar{x}}{S_{xx}} + \frac{x_0^2}{S_{xx}} \right)$, which rearranges to Eq. 5.5.2. Finally $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$ (the line passes through the point of means), and $\text{Cov}(\hat{\beta}_0 + \hat{\beta}_1 \bar{x}, \hat{\beta}_1) = -\sigma^2 \bar{x}/S_{xx} + \bar{x} \sigma^2/S_{xx} = 0$. ■

Three design lessons can be read off. The slope is estimated precisely when the x_i are spread out, since its variance is inversely proportional to S_{xx} . The intercept is estimated precisely only near the data: its variance grows with $\bar{x}^2$, and the estimated intercept and slope are strongly negatively correlated when $\bar{x}$ is large and positive, the tilting of the ellipses noted after Theorem 5.2.1. And the mean response is estimated best at $x_0 = \bar{x}$, with a variance that grows quadratically as x_0 moves away. Extrapolation is doubly risky: the model may fail outside the range of the data, and even if it holds, the estimate is imprecise there.

Example. Where to put the design points

Suppose $n = 20$ observations can be taken anywhere in $[0, 10]$ to estimate a slope. Three designs are compared in Figure 5.5.1. Equally spaced points give $S_{xx} = 184.21$ and a slope standard deviation of 0.0737σ . Putting half the points at each end gives the largest possible $S_{xx} = 500.00$ (Exercise (B1)) and 0.0447σ . Bunching the points in the middle, over $[4, 6]$, gives $S_{xx} = 7.37$ and 0.3684σ , more than eight times worse than the two-ends design. The simulated sampling distributions in panel (b), each from 20,000 samples, match these standard deviations. The two-ends design has a price the variance formula does not show: with only two distinct values of x , the data cannot reveal curvature. The equally spaced design gives up some precision for the ability to check the model.

```
import numpy as np
n = 20
designs = {
    "equally spaced": np.linspace(0, 10, n),
    "two ends": np.repeat([0.0, 10.0], n // 2),
```

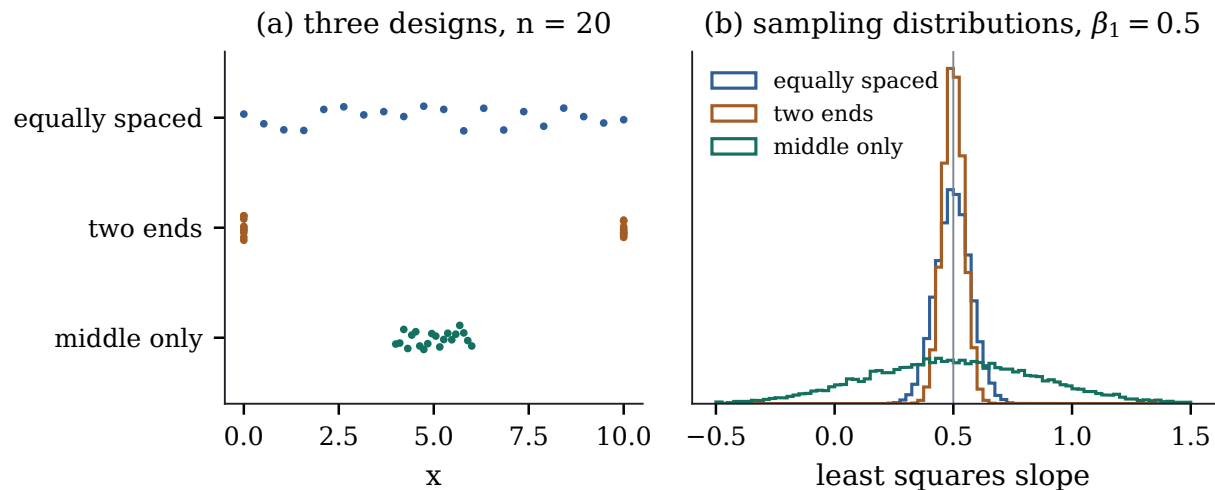


Figure 5.5.1: (a) Three designs with 20 points in $[0, 10]$. (b) Sampling distributions of the least squares slope under each design, with $\beta_1 = 0.5$ and $\sigma = 1$. All three are centred on β_1 ; their spreads are proportional to $S_{xx}^{-1/2}$.

```

    "middle only": np.linspace(4, 6, n),
}
for name, xd in designs.items():
    Sxx = np.sum((xd - xd.mean()) ** 2)
    print(f"{name:15s} S_xx = {Sxx:7.2f}    sd(slope) = sigma * {1 / np.sqrt(Sxx):.4f}
    ")

```

[Run this code online](#)

5.5.4 Only two moments are needed

Example. Skewed errors

To see that normality plays no role in [Theorem 5.5.1](#), fit the quadratic $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$ at $x = 1, \dots, 12$, with $\beta = (2, 0.5, -0.03)^\top$ and errors that are centred exponential variables scaled to $\sigma = 1.5$. These errors are strongly skewed. Over 100,000 simulated data sets, the averages of the three estimates are 1.9974, 0.5005 and -0.0300 , and the simulated standard deviations are 1.5421, 0.5457 and 0.0409, against 1.5503, 0.5483 and 0.0411 from [Eq. 5.5.1](#). Every entry of the simulated covariance matrix is within 0.011 of the formula, measured as a correlation. The formula also predicts that $\hat{\beta}_1$ and $\hat{\beta}_2$ have correlation -0.973 , because the columns x and x^2 are nearly collinear on $1, \dots, 12$. A slightly larger linear coefficient can be almost exactly compensated by a slightly more negative quadratic one.

```

import numpy as np
rng = np.random.default_rng(5)
x = np.arange(1.0, 13.0) # twelve design points
X = np.column_stack([np.ones_like(x), x, x**2])
beta = np.array([2.0, 0.5, -0.03])
sigma = 1.5
reps = 100_000

A = np.linalg.solve(X.T @ X, X.T) # beta_hat = A y
E = sigma * (rng.exponential(size=(reps, len(x))) - 1.0) # skewed, mean 0, sd

```

```

sigma
B = (X @ beta + E) @ A.T # one estimate per row

print("mean of estimates:", np.round(B.mean(axis=0), 4), " true:", beta)
print("Monte Carlo covariance:\n", np.round(np.cov(B.T), 5))
print("sigma^2 (X^T X)^{-1}:\n", np.round(sigma**2 * np.linalg.inv(X.T @ X), 5))

```

[Run this code online](#)

5.5.5 Random regressors

When the regressors are random, the assumptions are read conditionally on $\mathbf{X}$ (Section 5.1). The conditional results then transfer.

Corollary 5.5.5. Random regressors

Let $(\mathbf{X}, \mathbf{Y})$ be random, with $\text{rank}(\mathbf{X}) = p$ with probability one, and suppose $E(\mathbf{Y} | \mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y} | \mathbf{X}) = \sigma^2 \mathbf{I}_n$. If $E[(\mathbf{X}^\top \mathbf{X})^{-1}]$ is finite, then

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}].$$

Proof. Conditionally on $\mathbf{X}$, Theorem 5.5.1 gives $E(\hat{\boldsymbol{\beta}} | \mathbf{X}) = \boldsymbol{\beta}$ and $\text{Cov}(\hat{\boldsymbol{\beta}} | \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$. By the laws of total expectation and covariance (Proposition 2.5.4), $E(\hat{\boldsymbol{\beta}}) = E(\boldsymbol{\beta}) = \boldsymbol{\beta}$ and $\text{Cov}(\hat{\boldsymbol{\beta}}) = E[\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}] + \text{Cov}(\boldsymbol{\beta}) = \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$, since $\boldsymbol{\beta}$ is constant. The finiteness assumption guarantees $\hat{\boldsymbol{\beta}}$ has finite second moments, which Proposition 2.5.4 needs. ■

This is what Example (One slope, two sampling schemes) showed by simulation: under the random design, the variance of the slope was $\sigma^2 E(1/S_{xx})$, not σ^2/S_{xx} for any one design. For inference, the conditional covariance $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ is the relevant one, because it describes the design actually observed.

5.5.6 Consistency

As the sample grows, a good estimator should approach the truth. For a fixed p and a sequence of model matrices $\mathbf{X}_n$, the mean squared error of $\hat{\boldsymbol{\beta}}$ is

$$E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \text{tr Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \text{tr}(\mathbf{X}_n^\top \mathbf{X}_n)^{-1},$$

by Theorem 5.5.1 and the fact that the trace of a covariance matrix is the sum of the variances.

Proposition 5.5.6. Consistency in mean square

Under the linear model with $\mathbf{X}_n$ of full rank p for every n , if $\text{tr}(\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \rightarrow 0$, equivalently if the smallest eigenvalue of $\mathbf{X}_n^\top \mathbf{X}_n$ tends to infinity, then $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \rightarrow 0$, and hence $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \delta) \rightarrow 0$ for every $\delta > 0$.

Proof. The first claim is the display above. The eigenvalues of $(\mathbf{X}_n^\top \mathbf{X}_n)^{-1}$ are the reciprocals $1/\lambda_1, \dots, 1/\lambda_p$ of those of $\mathbf{X}_n^\top \mathbf{X}_n$, all positive, so the trace is $\sum_j 1/\lambda_j$. It tends to zero iff every $\lambda_j \rightarrow \infty$, iff the smallest does. The last claim is Markov's inequality applied to $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2$: $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \delta) \leq E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2/\delta^2$. ■

For a straight line the condition is $S_{xx} \rightarrow \infty$ together with $\bar{x}^2/S_{xx} \rightarrow 0$, which holds for any design that keeps spreading, but fails, for instance, for $x_i = 1/i$, where S_{xx} converges to a finite limit. Then even infinitely many observations do not pin down the slope ([Exercise \(C1\)](#)).

5.5.7 What if the assumptions fail?

The proof of [Theorem 5.5.1](#) shows exactly what each assumption contributes, so it also shows what goes wrong without it. Two cases recur throughout the book.

Proposition 5.5.7. Moments under departures

Let $\mathbf{X}$ have full rank, $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$.

- (a) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$ for some $\boldsymbol{\delta} \in \mathbb{R}^n$, then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\delta}$.
- (b) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$ for some covariance matrix $\boldsymbol{\Sigma}$, then $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ and

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (5.5.3)$$

Proof. Both are Theorem 2.2.1 applied to $\mathbf{A}\mathbf{Y}$: $E(\mathbf{A}\mathbf{Y}) = \mathbf{A}\mathbf{X}\boldsymbol{\beta} + \mathbf{A}\boldsymbol{\delta}$ and $\text{Cov}(\mathbf{A}\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$. ■

Part (a) says that when the mean is misspecified, least squares estimates not $\boldsymbol{\beta}$ but $\boldsymbol{\beta}$ plus the coefficients from regressing the missing part $\boldsymbol{\delta}$ on $\mathbf{X}$. The most important case is an omitted regressor, which [Section 5.7](#) examines. Part (b) says that with the wrong covariance, the estimator stays unbiased but the formula $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ is wrong. The “sandwich” [Eq. 5.5.3](#) is the correct covariance, and [Chapter 21](#) shows how to estimate it when $\boldsymbol{\Sigma}$ is diagonal but unknown (Theorem 21.4.2). [Chapter 19](#) studies both departures systematically (Theorem 19.1.2, Theorem 19.2.2).

5.5.8 Exercises

5.5.8.1 A. Check your understanding

Exercise. A1

In the model of Proposition 5.4.4 with $c = 2$ groups, show that $\text{Var}(\hat{\beta}_1) = \sigma^2(1/n_1 + 1/n_2)$, both from [Eq. 5.5.1](#) and directly from $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_2$. For fixed $n = n_1 + n_2$, which split minimizes it?

Exercise. A2

Show that for a straight line $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$. Which case in Example (Forecasting December sea temperature) has the largest leverage? Check that the leverages sum to 2.

5.5.8.2 B. Practice

Exercise. B1

Suppose n is even and the x_i may be chosen anywhere in $[a, b]$. Show that $S_{xx} \leq n(b-a)^2/4$, with equality iff half the points are at a and half at b . *Hint:* $|x_i - c| \leq (b-a)/2$ with $c = (a+b)/2$, and $S_{xx} = \sum_i (x_i - c)^2 - n(\bar{x} - c)^2$.

Solution. With c the midpoint, $S_{xx} = \sum_i (x_i - c)^2 - n(\bar{x} - c)^2 \leq \sum_i (x_i - c)^2 \leq n(b-a)^2/4$. Equality in the second step needs every $x_i \in \{a, b\}$, and in the first step needs $\bar{x} = c$, that is, equally many points at each end. For $[0, 10]$ and $n = 20$ this gives $S_{xx} = 20 \cdot 25 = 500$, the value in [Example \(Where to put the design points\)](#).

Exercise. B2

In Example (Money demand in Denmark), write the variances of $\hat{\beta}_2 - \hat{\beta}_3$ and $\hat{\beta}_2 + \hat{\beta}_3$, the difference and the sum of the two interest-rate coefficients, in terms of the entries of $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^{-1}$. Show that if $\hat{\beta}_2$ and $\hat{\beta}_3$ are negatively correlated, the difference is estimated *less* precisely, and the sum *more* precisely, than they would be if the two estimates were uncorrelated. The two rate columns have sample correlation 0.80. Which combination would you expect to be better determined?

Solution. With $\mathbf{a} = (0, 0, 1, \mp 1)^\top$ and indices counting the intercept as the first coefficient, [Corollary 5.5.2](#) gives $\text{Var}(\hat{\beta}_2 \mp \hat{\beta}_3) = \sigma^2(g_{33} + g_{44} \mp 2g_{34})$, and $\text{Cov}(\hat{\beta}_2, \hat{\beta}_3) = \sigma^2 g_{34}$. If $g_{34} < 0$, the variance of the difference exceeds $\sigma^2(g_{33} + g_{44}) = \text{Var}(\hat{\beta}_2) + \text{Var}(\hat{\beta}_3)$, and that of the sum falls below it. Positively correlated columns tend to produce negatively correlated estimates (as in [Example \(Skewed errors\)](#)), because the data can tell how large the combined effect of the two rates is more easily than how to divide it between them. So the sum should be the better determined combination.

5.5.8.3 C. Going deeper

Exercise. C1

Let $x_i = 1/i$ for $i = 1, 2, \dots$ in the straight-line model. Show that S_{xx} converges to a finite limit as $n \rightarrow \infty$, so that $\text{Var}(\hat{\beta}_1)$ does not tend to zero. Then construct a sequence of designs, in which the points may change with n (for example, designs centred farther and farther from 0), with $S_{xx} \rightarrow \infty$ but $\bar{x}^2/S_{xx} \not\rightarrow 0$, and decide whether the slope, the intercept, or both fail to be consistent.

Solution. With $x_i = 1/i$, $S_{xx} = \sum_{i \leq n} 1/i^2 - n\bar{x}^2$. The sum increases to $\pi^2/6$, and $n\bar{x}^2 = H_n^2/n \rightarrow 0$ because the harmonic number $H_n = \sum_{i \leq n} 1/i$ grows like $\log n$. So $S_{xx} \rightarrow \pi^2/6$ and $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$ stays above $6\sigma^2/\pi^2$.

For even n take $x_i = n + (-1)^i$, $i = 1, \dots, n$: half the points at $n - 1$, half at $n + 1$. Then $\bar{x} = n$ and $S_{xx} = n$, so $\bar{x}^2/S_{xx} = n \rightarrow \infty$. By [Corollary 5.5.4](#), $\text{Var}(\hat{\beta}_1) = \sigma^2/n \rightarrow 0$, so the slope is consistent (Chebyshev's inequality), while $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + n) \rightarrow \infty$. The intercept, the mean response at $x = 0$, ever farther from the data, is not consistent: with normal errors, $\hat{\beta}_0 - \beta_0$ is normal with unbounded variance.